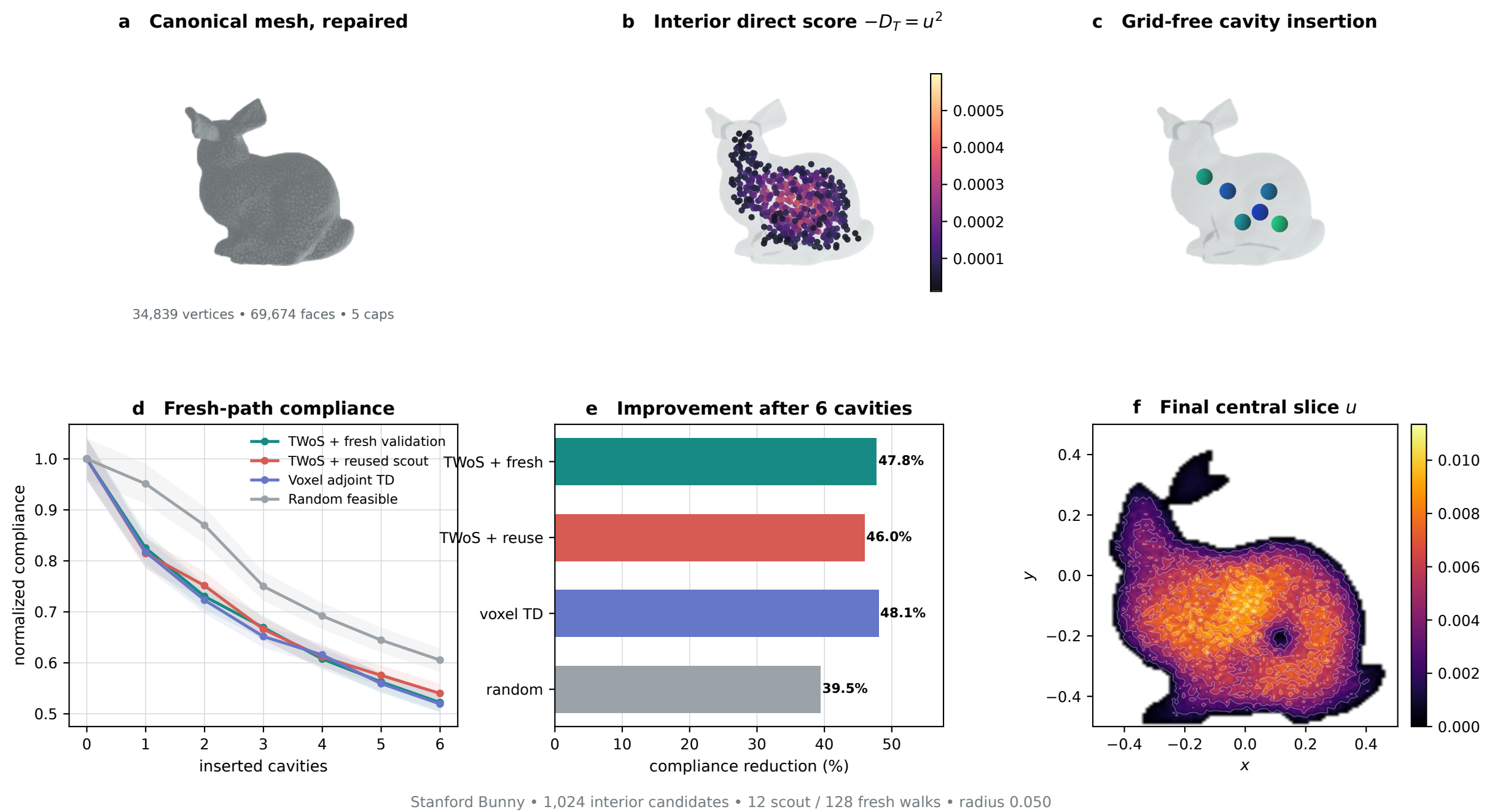

a Canonical mesh, repaired
34,839 vertices • 69,674 faces • 5 caps
b Interior direct score $-D_T = u^2$
0.0005
0.0004
0.0003
0.0002
0.0001
c Grid-free cavity insertion
d Fresh-path compliance
TWoS + fresh validation
TWoS + reused scout
Voxel adjoint TD
Random feasible
normalized compliance
inserted cavities
e Improvement after 6 cavities
TWoS + fresh
TWoS + reuse
voxel TD
random
47.8%
46.0%
48.1%
39.5%
compliance reduction (%)
f Final central slice u
x
y
Stanford Bunny • 1,024 interior candidates • 12 scout / 128 fresh walks • radius 0.050